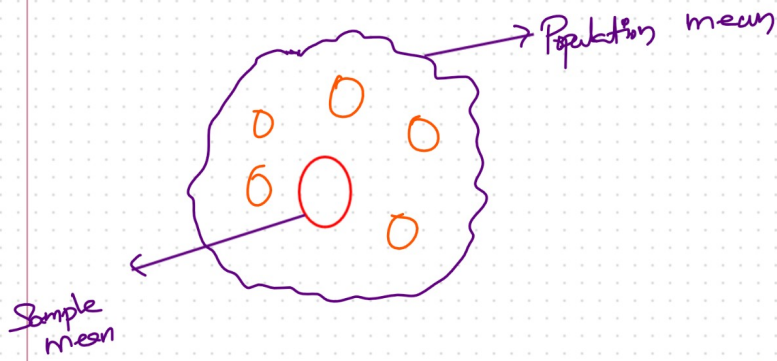


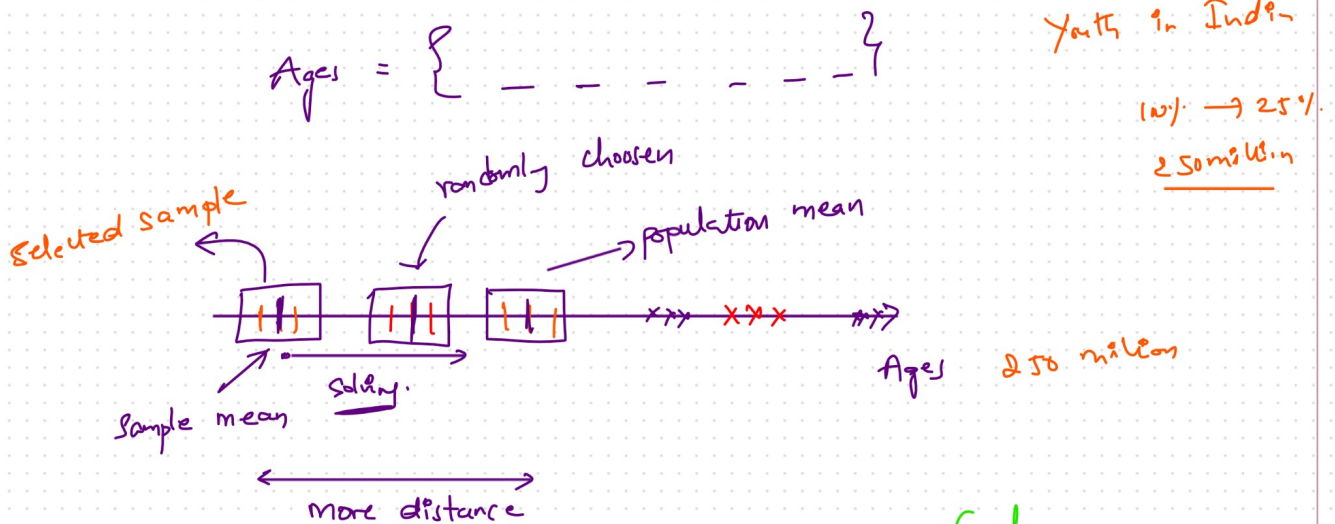
Statistics



$$s^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1} \rightarrow \text{why not } n?$$

→ we take sample data to infer population data after various experiments

→ So here mean calculated on sample data might not be the same $\therefore$ which leads in under-estimation of population variance



$$\therefore \bar{x} \approx \mu \quad \text{Goal}$$

$$s^2 \approx \sigma^2 \quad \text{when chosen randomly}$$

there is a chance for example the first set of sample mean which is bit far from population mean

$$\therefore \bar{x} \ll \mu$$

$$s^2 \ll \sigma^2$$

$\therefore$ Now we are under-estimating the true population variance with this sample data

∴ when we use $(n-1)$ which is a smallest number to come near the population mean

This entire scenario is called as Bessel Correction

↓
also termed as degree of freedom

$$\boxed{dof = n-1}$$

Percentiles & Quartiles :

w.r.t percentage { 1, 2, 3, 4, 5, 6 }

No. of odd numbers = 3

$$\% \text{ of odd numbers} = \frac{3}{6} \times 100 = 50\%$$

Percentile: A percentile is a value below which certain percentage of observations lie

{ 2, 2, 3, 4, 5, 5, 6, 7, 8, 8, 8, 9, 9, 10 }

↑ ↑

$n = 14$

$$\text{Percentile of value } x = \frac{\# \text{ no. of values below } x}{n} \times 100$$

$$= \frac{11}{14} \times 100$$

Percentile ranking = 78.57 % of value 9

So here 78.57% of values from entire / total distribution is less than 9

Calculate value from percentile:

$$\text{Value} = \frac{\text{Percentile}}{100} * (n+1)$$

$p=25\%$

$$= \frac{25}{100} \times 15$$
$$= 3.75$$

{ 2, 2, 3, 4, 5, 5, ... }

$$\frac{3+4}{2} = 3.5$$

25% of value is 3.5

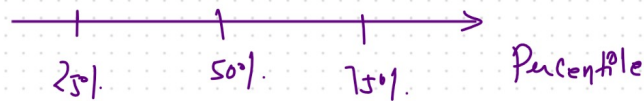
Quartiles

25% = 1st Quartile

50% = 2nd Quartile

75% = 3rd Quartile

}



5 Number Summary

In ML, it helps us to feature selection & how to remove outliers & also helps to create plots

① Minimum

② 1st Quartile (25%)

③ Median

④ 3rd Quartile (75%)

⑤ Maximum

ex:

$$X = \{1, 2, 2, 2, 3, 3, 4, 5, 5, 5, 6, 6, 6, 6, 7, 8, 8, 9, 27\}$$

[lower fence $\longleftrightarrow$ higher fence]

$$\rightarrow \text{lower fence} = Q_1 - 1.5 (IQR)$$

$$\rightarrow \text{Higher Fence} = Q_3 + 1.5 (IQR)$$

$$IQR = Q_3 - Q_1$$

Intu Quantile Range

fence method

$\therefore$ numbers less than lower fence & higher than higher fence we consider them as outliers

Pb Solution

$$Q_1 = \frac{\text{Percentile}}{100} \times (n+1)$$

(25%)

$$= \frac{25}{100} \times 20 = 5^{\text{th}} \text{ position} \Rightarrow 3$$

$$(75\%) Q_3 = \frac{75}{100} \times 20 = 15^{\text{th}} \text{ position} = 7$$

$$IQR = Q_3 - Q_1 = 7 - 3 = 4$$

$\therefore$ Calculate lower & higher fence

$$\text{lower fence} = Q_1 - 1.5 (IQR)$$

$$= 3 - 1.5 (4)$$

$$= 3 - 6$$

$$= -3$$

$$\begin{aligned}
 \text{Higher fence} &= Q_3 + 1.5(IQR) \\
 &= 7 + 1.5(4) \\
 &= 7 + 6 \\
 &= 13
 \end{aligned}$$

$$< [-3 \longleftrightarrow 13] > \text{ outliers}$$

$$\begin{aligned}
 X &= \{ 1, 2, 2, 2, 3, 3, 4, 5, 5, 5, 6, 6, 6, 6, 7, 8, 8, 9, 27 \} \\
 &\quad \downarrow \\
 &= \{ 1, 2, 2, 2, 3, 3, 4, 5, 5, 5, 6, 6, 6, 6, 7, 8, 8, 9 \}
 \end{aligned}$$

$\therefore$ Now 5 number summary will be

$$\text{minimum} = 1$$

$$\text{maximum} = 9$$

$\therefore$ 27 is removed cause outlier

$$1^{\text{st}} \text{ Quartile} = 3$$

$$\text{median} = 5$$

$$3^{\text{rd}} \text{ Quartile} = 7$$

Q You are a data scientist at an e-commerce company. The daily transactions values in (thousand ₹) for the past 10 days sorted

$$1.2, 2.5, 3.1, 3.8, 4.2, 4.9, 5.3, 6.1, 7.0, 8.5$$

Tukey Fence method

